BAO CAO THUC NGHIEM LAB01

Run ID: 20261104_230429_520fe2
Trang thai: SUCCESS
Thoi gian bat dau: 2026-11-04T23:04:29+07:00
Thoi gian ket thuc: 2026-11-04T23:05:22+07:00

1. TIEN TRINH THUC HIEN

STT	Buoc thuc hien	Trang thai
1	EDA	SUCCESS
2	PREPROCESSING	SUCCESS
3	TRAINING - Logistic Regression	SUCCESS
4	TRAINING - Naive Bayes	SUCCESS
5	TRAINING - SVM	SUCCESS
6	TRAINING	SUCCESS
7	EVALUATION - Logistic Regression - Train	SUCCESS
8	EVALUATION - Logistic Regression - Test	SUCCESS
9	EVALUATION - Naive Bayes - Train	SUCCESS
10	EVALUATION - Naive Bayes - Test	SUCCESS
11	EVALUATION - SVM - Train	SUCCESS
12	EVALUATION - SVM - Test	SUCCESS
13	EVALUATION	SUCCESS
14	PIPELINE	SUCCESS

2. KET QUA DANH GIA MO HINH

Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
Logistic Regression	Train	0.9535472972972973	0.9007832898172323	0.9530386740331491	0.9261744966442953	0.9530386740331491	0.046228710462287104	0.9716361438211613	3136	152	68	1380
Logistic Regression	Test	0.9315878378378378	0.8469135802469135	0.9475138121546961	0.8943937418513689	0.9475138121546961	0.075425790754257915	0.9596792622763506	760	62	19	343
Naive Bayes	Train	0.9961993243243243	0.9897260273972602	0.9979281767955801	0.9938101788170564	0.9979281767955801	0.004562043795620438	0.9999590424244869	3273	15	3	1445
Naive Bayes	Test	0.9729729729729729	0.9634831460674157	0.9475138121546961	0.9554317548746517	0.9475138121546961	0.015815085158150853	0.9935476065653103	809	13	19	343
SVM	Train	0.7193834459459459	0.8888888888888888	0.09392265193370165	0.16989381636477202	0.09392265193370165	0.005170316301703163	0.9747585813472059	3271	17	1312	136

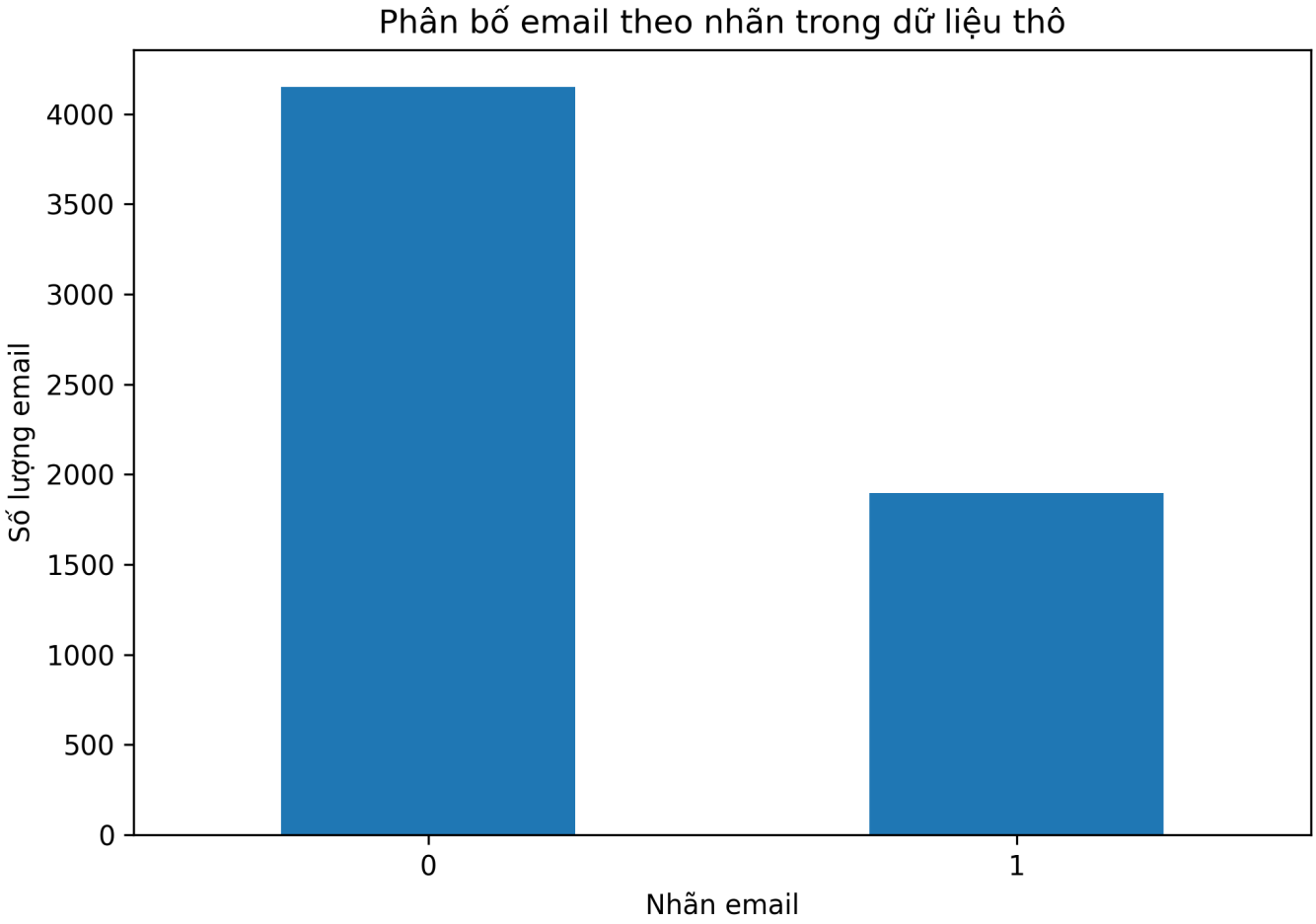
Mô hình	T■p d■ li■u	Accuracy	Precision	Recall	F1	TPR	FPR	AUC	TN	FP	FN	TP
SVM	Test	0.71875	0.8918918918918919	0.09116022099447514	0.16541353383458646	0.09116022099447514	0.004866180048661801	0.9647269158903631	818	4	329	33

3. MO HINH TOT NHAT

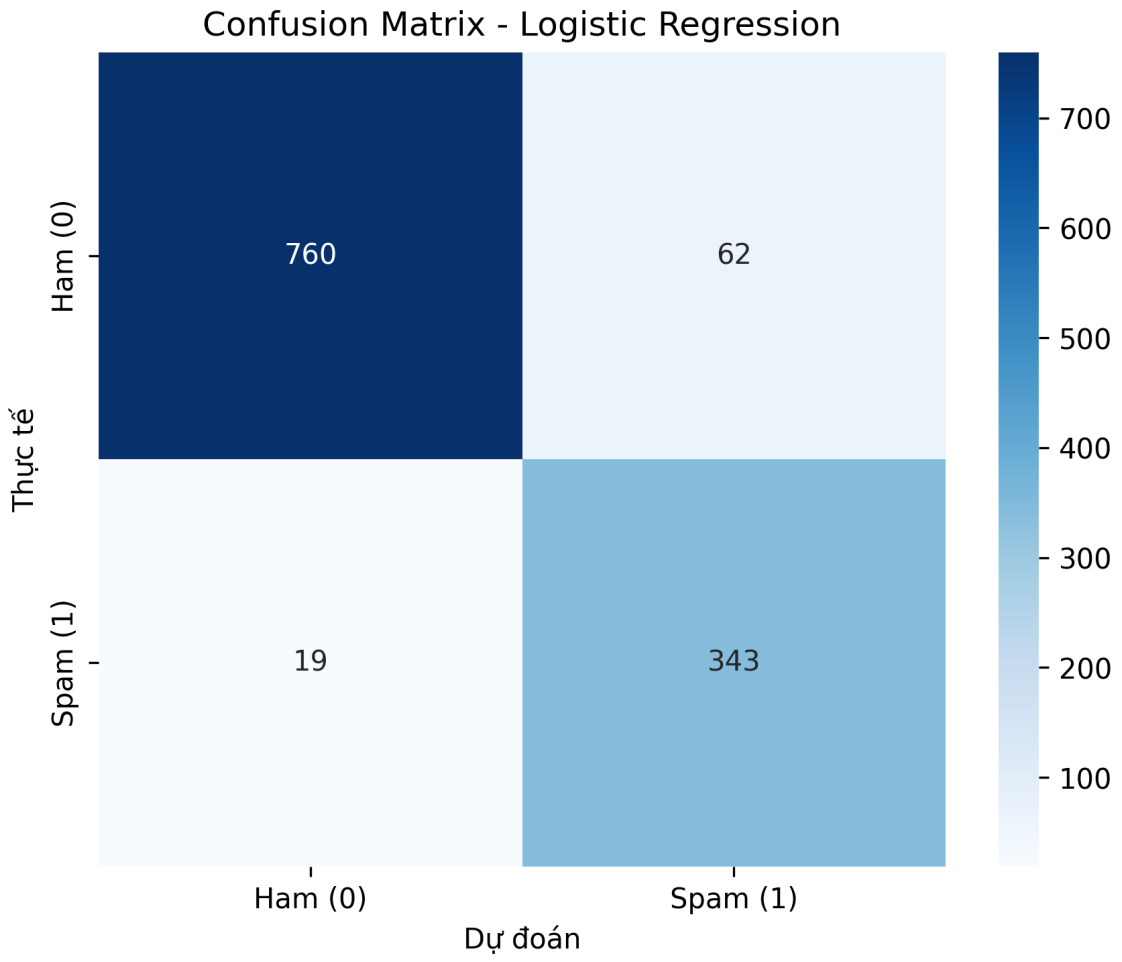
Mo hinh: Naive Bayes
F1-score: 0.9554317548746517

4. BIEU DO KET QUA

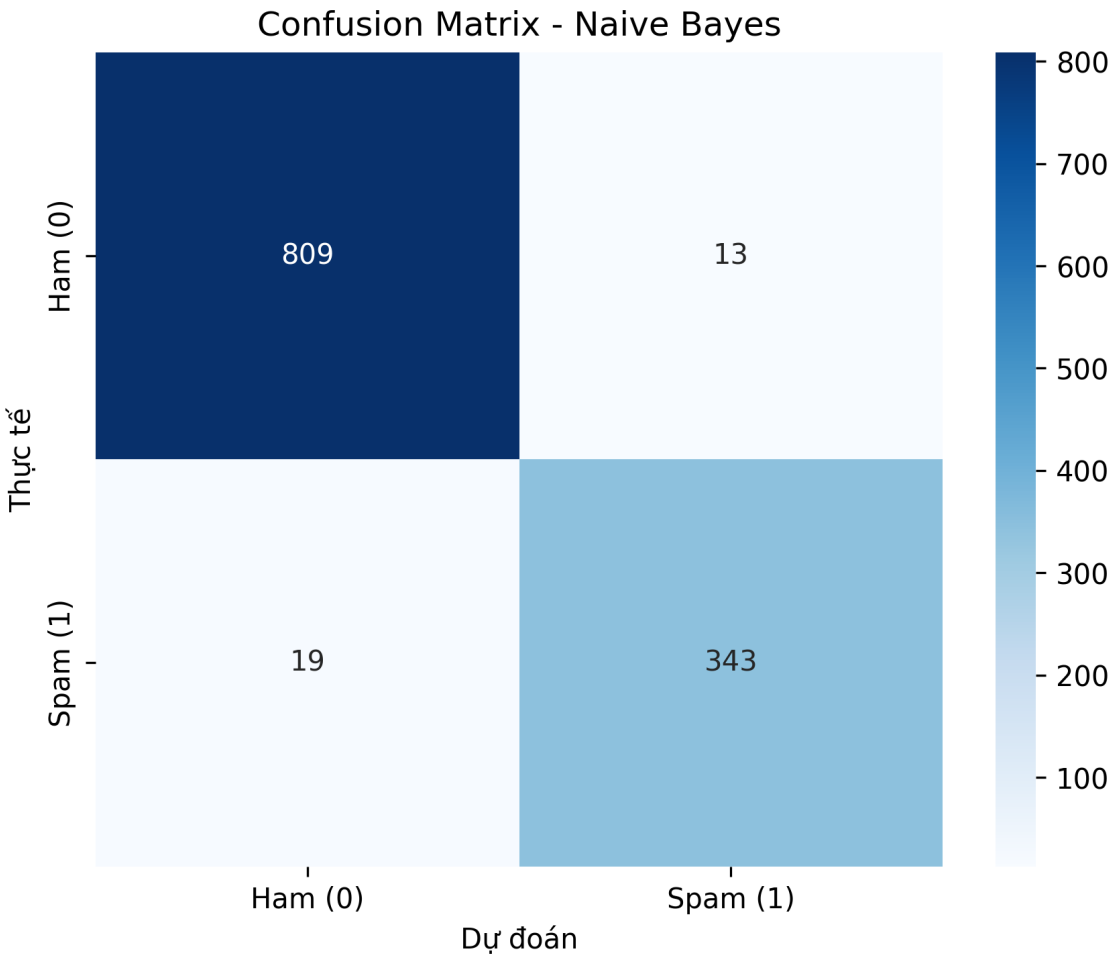
label_distribution



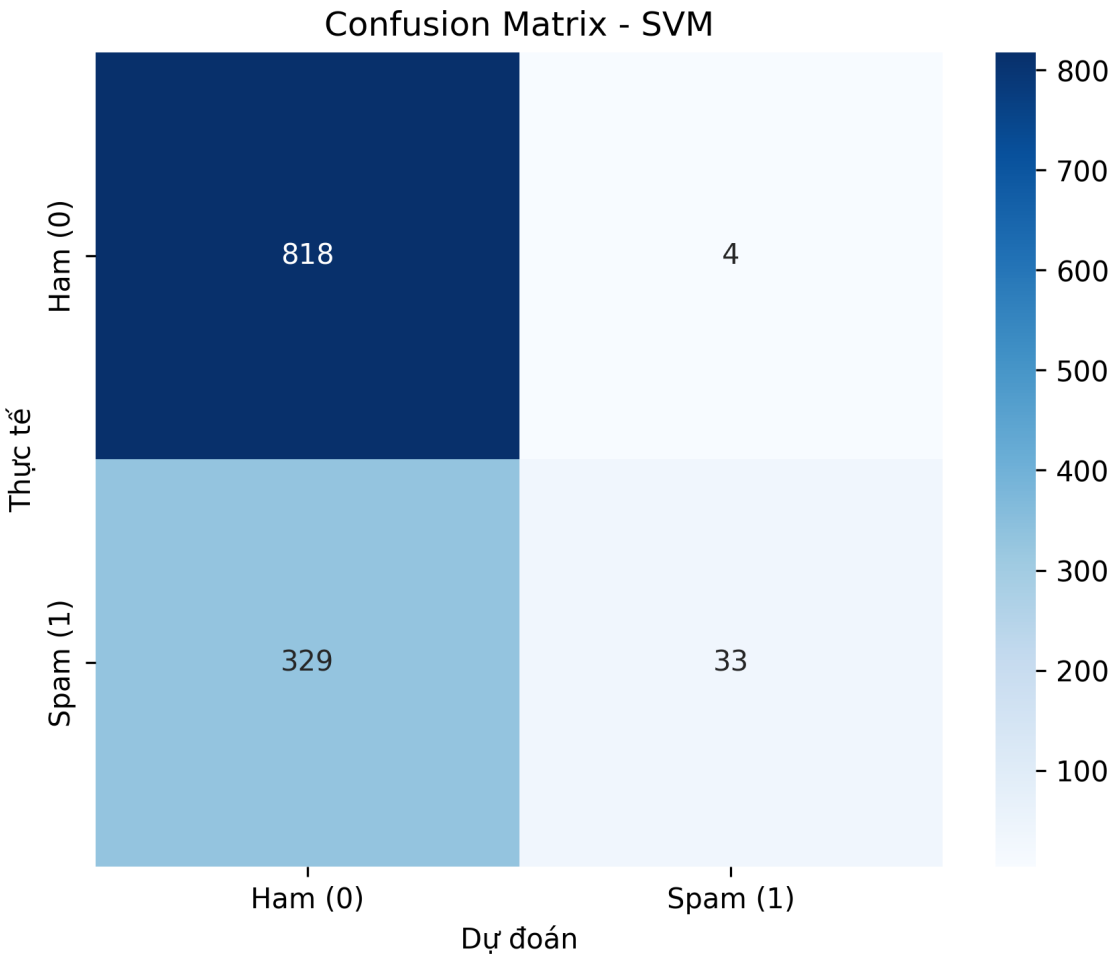
confusion_matrix_lr



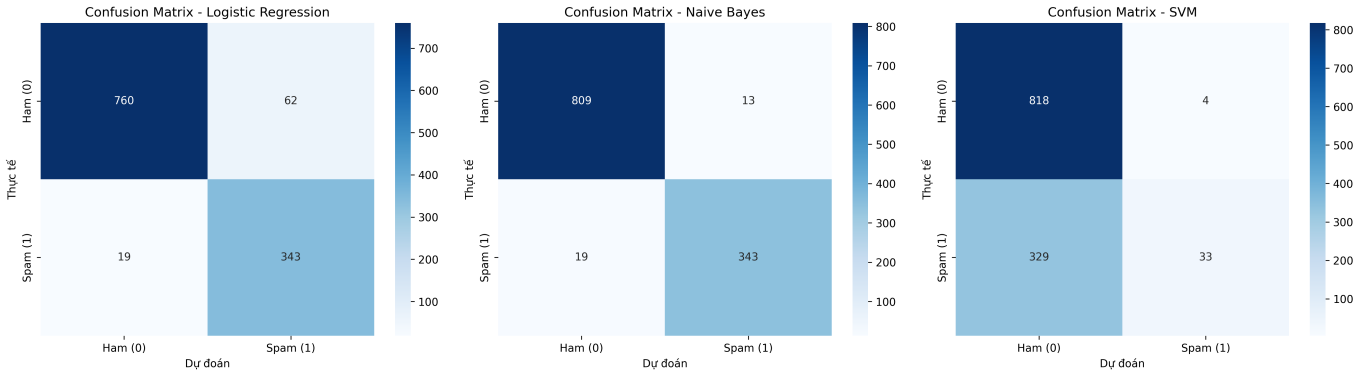
confusion_matrix_nb



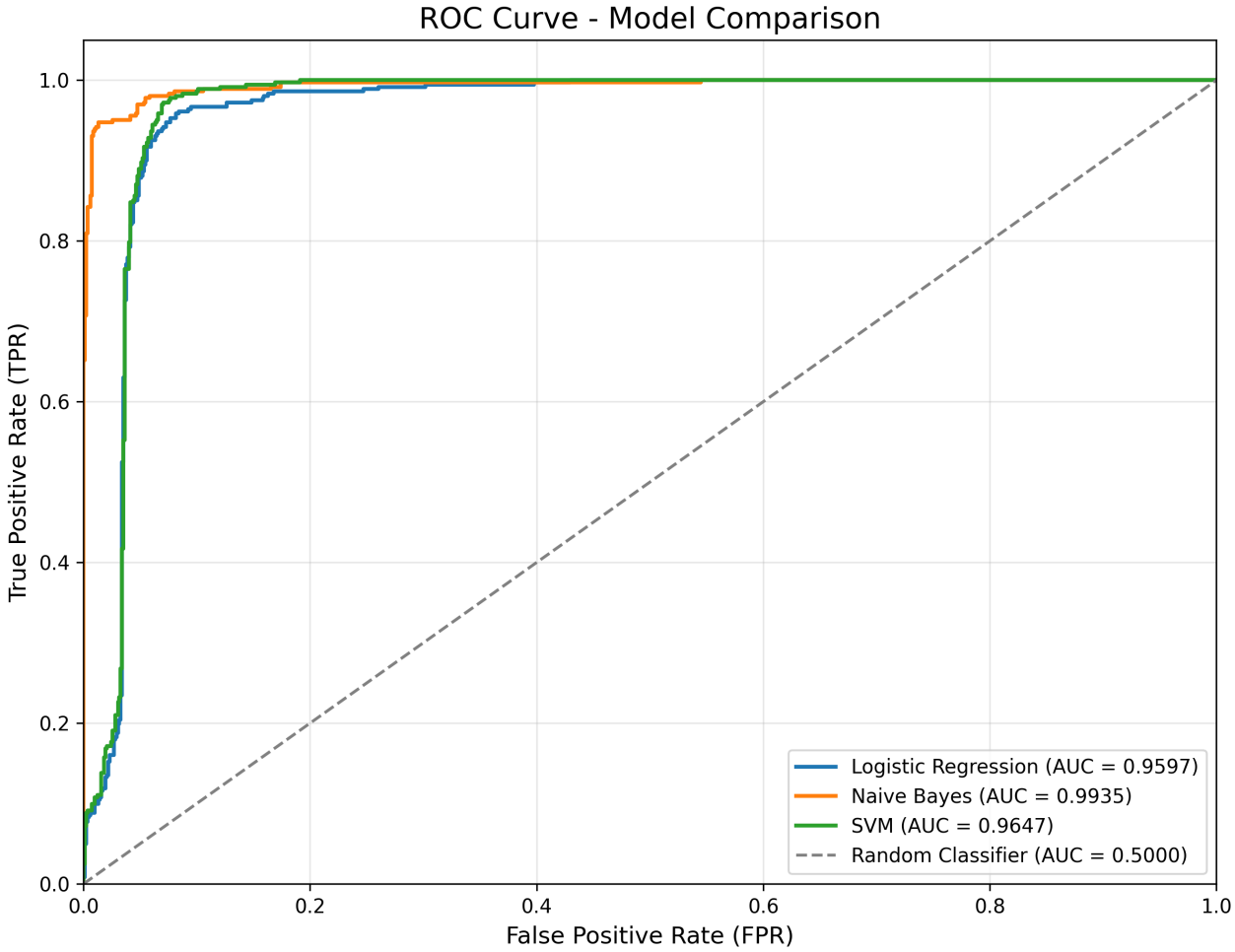
confusion_matrix_svm



confusion_matrix_all



roc_curve_all


ROC Curve - Model Comparison
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.9597)
Naive Bayes (AUC = 0.9935)
SVM (AUC = 0.9647)
Random Classifier (AUC = 0.5000)

5. NHAT KY THUC THI

[2026-11-04T23:04:29+07:00] [START] [PIPELINE] Bat dau quy trinh may hoc.

[2026-11-04T23:04:29+07:00] [START] [EDA] Bat dau buoc EDA.

[2026-11-04T23:04:29+07:00] [START] [EDA] Bat dau phan tich du lieu

[2026-11-04T23:04:29+07:00] [SUCCESS] [EDA] Hoan thanh EDA trong 0.233 giay

[2026-11-04T23:04:29+07:00] [SUCCESS] [EDA] Hoan thanh trong 0.25 giay.

[2026-11-04T23:04:29+07:00] [START] [PREPROCESSING] Bat dau buoc PREPROCESSING.

[2026-11-04T23:04:29+07:00] [START] [PREPROCESSING] Bat dau tien xu ly du lieu

[2026-11-04T23:04:31+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh tien xu ly trong 2.3143 giay

[2026-11-04T23:04:31+07:00] [SUCCESS] [PREPROCESSING] Hoan thanh trong 2.34 giay.

[2026-11-04T23:04:31+07:00] [START] [TRAINING] Bat dau buoc TRAINING.

[2026-11-04T23:04:31+07:00] [START] [TRAINING] Bat dau huan luyen mo hinh

[2026-11-04T23:05:12+07:00] [SUCCESS] [TRAINING - Logistic Regression] Training completed in 40.62 seconds

[2026-11-04T23:05:12+07:00] [SUCCESS] [TRAINING - Naive Bayes] Training completed in 0.28 seconds

[2026-11-04T23:05:20+07:00] [SUCCESS] [TRAINING - SVM] Training completed in 7.48 seconds

[2026-11-04T23:05:20+07:00] [SUCCESS] [TRAINING] Hoan thanh training trong 48.603 giay

[2026-11-04T23:05:20+07:00] [SUCCESS] [TRAINING] Hoan thanh trong 48.65 giay.

[2026-11-04T23:05:20+07:00] [START] [EVALUATION] Bat dau buoc EVALUATION.

[2026-11-04T23:05:20+07:00] [START] [EVALUATION] Bat dau danh gia cac mo hinh.

[2026-11-04T23:05:20+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Train] Danh gia Logistic Regression tren Train hoan tat. F1=0.9262, AUC=0.9716

[2026-11-04T23:05:20+07:00] [SUCCESS] [EVALUATION - Logistic Regression - Test] Danh gia Logistic Regression tren Test hoan tat. F1=0.8944, AUC=0.9597

[2026-11-04T23:05:21+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Train] Danh gia Naive Bayes tren Train hoan tat. F1=0.9938, AUC=1.0000

[2026-11-04T23:05:21+07:00] [SUCCESS] [EVALUATION - Naive Bayes - Test] Danh gia Naive Bayes tren Test hoan tat. F1=0.9554, AUC=0.9935

[2026-11-04T23:05:21+07:00] [SUCCESS] [EVALUATION - SVM - Train] Danh gia SVM tren Train hoan tat. F1=0.1699, AUC=0.9748

[2026-11-04T23:05:21+07:00] [SUCCESS] [EVALUATION - SVM - Test] Danh gia SVM tren Test hoan tat. F1=0.1654, AUC=0.9647

[2026-11-04T23:05:22+07:00] [SUCCESS] [EVALUATION] Danh gia hoan tat trong 1.93 giay.

[2026-11-04T23:05:22+07:00] [SUCCESS] [EVALUATION] Hoan thanh trong 1.95 giay.

[2026-11-04T23:05:22+07:00] [SUCCESS] [PIPELINE] Hoan thanh toan bo quy trinh trong 53.19 giay.